
Clustering and automatic segmentation of natural images using deep learning and unsupervised techniques

Volpellier Daniela, Terán Guillermo, Turri Valentín and Sandagorda Jessica

Master's Program in Data Mining and Knowledge Discovery

University of Buenos Aires

Buenos Aires

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Abstract

This work analyzes the application of unsupervised clustering techniques to a multiclass image dataset, using features extracted with VGG16. K-Means, DBSCAN (with different metrics), and hierarchical clustering were applied and evaluated using internal and external metrics. In addition, a complementary task of automatic segmentation on a selected image was developed using Connected Component Labelling and spectral clustering. The results show that the choice of algorithm and the representation space significantly impact clustering quality.

1 Introduction

Automatic image analysis is a technique used to process large volumes of visual data that cannot be manually evaluated. This type of information is generated by sensors, cameras, telescopes, or biomedical equipment, and requires computational methods for organization and analysis (Aggarwal, 2015).

Unsupervised learning, particularly clustering, makes it possible to divide objects into groups according to their similarity without the need for prior labels. This task is based on defining a distance metric and maximizing intra-cluster cohesion and inter-cluster separation (Tan, Steinbach, & Kumar, 2018). There are multiple algorithms, including k-means, DBSCAN, and hierarchical methods, with different behaviors depending on the shape and density of the data (Jain, 2010).

Image processing involves handling high-dimensional vectors, which produces effects such as data dispersion and loss of contrast between observations. This situation is known as the “curse of dimensionality” and affects the effectiveness of clustering algorithms (Bengio, Courville, & Vincent, 2013). To mitigate it, dimensionality reduction or feature extraction via convolutional neural networks is commonly used. The VGG16 model is a feature extractor based on transfer learning, trained on the ImageNet dataset and widely used to obtain numerical image representations (Simonyan & Zisserman, 2015).

The evaluation of the output of a clustering algorithm can be performed through internal or external validation. Internal validation includes metrics such as the silhouette coefficient (Rousseeuw, 1987) and the within-cluster sum of squared errors (SSE). External validation compares the obtained clusters with reference labels using measures such as the Adjusted Rand Index (Hubert & Arabie, 1985) and Normalized Mutual Information (Manning, Raghavan, & Schütze, 2008).

The objective of this work is to apply clustering algorithms to VGG16-extracted features on an image dataset, evaluate the performance of the methods using internal and external validation metrics, and analyze the coherence between the obtained clusters and the true classes of the dataset.

2 General methods

The dataset used was the Natural Images Dataset. It consists of 6,899 JPG images distributed across eight balanced classes, stored in separate folders by category (for example, /cat, /dog, etc.). The images are RGB color images and have variable original sizes, so they were resized to 224×224 pixels to ensure uniformity during processing. This step also included pixel-value normalization to match the requirements of the VGG16 model.

During the preparation stage, images and their labels were loaded, and their quality and structural compatibility were verified. Relevant dataset properties were documented, ensuring that images were comparable in terms of color, scale, resolution, and proportions. This step was essential to reduce potential biases and to ensure that subsequent models operate on a consistent dataset.

Next, the images were processed with the pretrained VGG16 model, removing the classification layers and using the output of the fully connected layer (fc1) as a feature vector. This representation produced 4,096-dimensional vectors capturing deep semantic properties of the images.

To facilitate both exploration and clustering, dimensionality reduction techniques were applied. PCA (Principal Component Analysis) was used for linear projections into two dimensions, and UMAP (Uniform Manifold Approximation and Projection) was used as a nonlinear technique aimed at preserving local neighborhoods.

On the original vectors and their reduced projections, three unsupervised clustering algorithms were applied: K-Means, DBSCAN (with Euclidean and Cosine distances), and agglomerative hierarchical clustering with ward linkage. In each case, the resulting clusters were compared with the true dataset labels, enabling an assessment of their coherence and quality.

Internal validation was performed using the silhouette coefficient and within-cluster sum of squared errors (SSE). External validation was carried out using the Adjusted Rand Index, Normalized Mutual Information, and the Van Dongen index, implemented manually. These metrics made it possible to quantify the correspondence between the generated clusters and the true classes of the dataset.

In addition, a complementary task of automatic object segmentation was addressed on a selected image. For this purpose, preprocessing techniques, connected component labeling, and a spectral clustering algorithm were applied using spatial and color features.

3 Experiment 1: K-Means

3.1 Specific methods

The K-Means algorithm was applied to the feature vectors extracted using the VGG16 network. Each image was represented by a 4096-dimensional vector corresponding to the output of the fc1 layer. To determine the optimal number of clusters (k), a systematic sweep was performed over a range of values between 2 and 99. For each value of k , two internal validation metrics were computed: SSE (Sum of Squared Errors), as the elbow-method criterion, and the silhouette coefficient, as a measure of intra-cluster cohesion and inter-cluster separation. Both metrics were plotted to select the most appropriate k .

Once a reasonable value was identified ($k = 7$), the K-Means model was run on the full dataset. The clustering results were compared with the true dataset labels using external validation metrics: the Adjusted Rand Index (ARI) and the Van Dongen index, the latter implemented manually from the confusion matrix.

3.2 Results and discussion

The K-Means analysis focused on evaluating the algorithm's ability to group images based on representations extracted from VGG16. Initially, k values from 2 to 99 were explored, computing for each partition the sum of squared errors (SSE) and the silhouette coefficient. The elbow-method curve showed a steep decrease for the first values and stabilization from $k=7$ onward, while the silhouette coefficient reached its maximum around the same point (silhouette = 0.237). These observations motivated the choice of $k=7$ for the subsequent stages of the analysis (SSE = 131,930).

Once the algorithm was applied with this value, each image was assigned to a cluster. To evaluate the coherence of these assignments with the original dataset categories (known in advance), two external validation metrics were computed. The Adjusted Rand Index (ARI), which measures the agreement between two partitions corrected for chance ($ARI = 0.8533$), and the Van Dongen index, which evaluates structural divergence between the clusters and the true classes (Van Dongen = 0.0686). Both indicators suggested partial correspondence: some clusters aligned clearly with certain classes (e.g., airplanes and cars), while others were heterogeneous, containing images from different categories.

In addition to these metrics, the clustering was visually analyzed using PCA dimensionality reduction. The two-dimensional plots (Figure 1) showed that, although there were dense areas corresponding to certain well-defined clusters, there were also regions of overlap between semantically similar classes, such as flowers and fruits, or cats and dogs. This indicates that the representation in the space generated by VGG16 does not always capture sharp separations between certain classes.

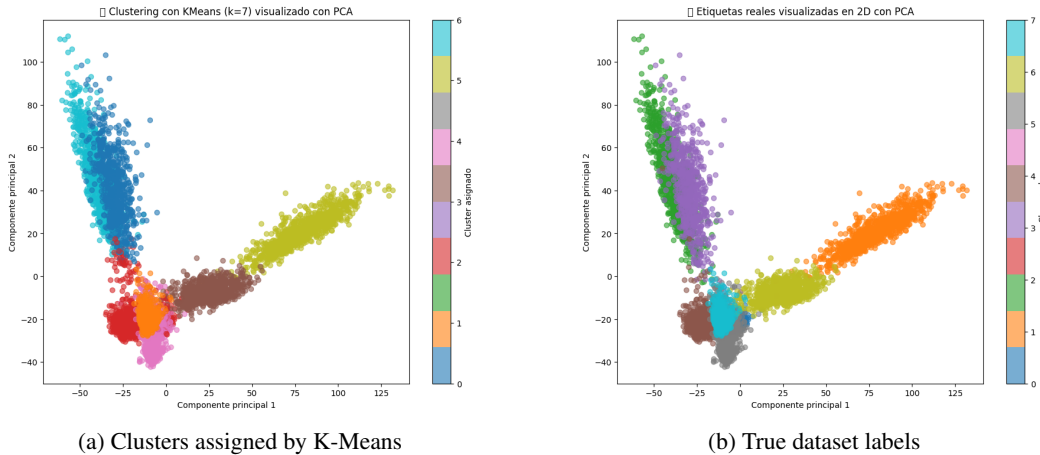


Figure 1: Two-dimensional visualization (PCA) comparing the clusters generated by K-Means (left) and the true dataset labels (right).

On the other hand, the confusion matrix (Figure 2) between true labels and assigned clusters was examined. It showed that some groups contained mostly a single class (indicating high purity), while others mixed several classes, evidencing structural dispersion in the feature space. This could be related both to the visual nature of the images and to limitations of the algorithm, which assumes the existence of spherical clusters of similar size—an assumption that is not always satisfied in practice.

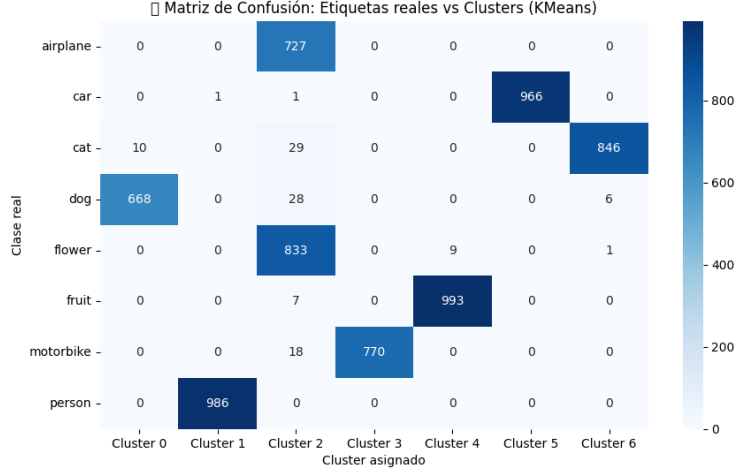


Figure 2: Confusion matrix between the true dataset labels and the clusters generated by K-Means. Each cell shows the number of images of a specific class (vertical axis) assigned to each cluster (horizontal axis).

In summary, the obtained results show that K-Means can capture a basic structure of the image dataset, achieving an acceptable segmentation in general terms. However, it has limitations when classes overlap visually or exhibit high internal variability. This suggests that, although it is a useful starting point, it may be convenient to complement it with methods that do not require specifying the number of clusters and that allow more flexible shapes. For this reason, DBSCAN is analyzed next.

4 Experiment 2: DBSCAN

4.1 Specific methods

The DBSCAN algorithm was applied to explore unsupervised clustering of images based on features extracted with VGG16. Unlike K-Means, DBSCAN does not require knowing the number of clusters in advance, but it depends on two key parameters: ϵ (maximum distance between points to be considered neighbors) and $\min_samples$ (minimum number of neighbors to form a core point). In all experiments, $\min_samples$ was fixed at 5.

To explore the algorithm’s behavior, three different approaches were applied. First, DBSCAN was run using Euclidean distance on the original vectors. Then, the procedure was repeated using cosine similarity as the distance metric, to evaluate how the chosen metric affects the shape and composition of the clusters. Finally, DBSCAN was applied to a representation space reduced with UMAP, a nonlinear dimensionality reduction technique that preserves local structure. This last variant aims to improve DBSCAN performance in high-dimensional spaces by reducing noise and geometric irrelevance.

For each variant, a systematic analysis of the ϵ parameter was performed, evaluating how the number of clusters and the silhouette coefficient vary. Configurations with $\epsilon = 20$ and $\epsilon = 15$ were selected as representative for a more detailed analysis. In each case, results were projected with PCA to visually inspect cluster distributions, and confusion matrices were built to compare the generated clusters with the true classes.

4.2 Results and discussion

The behavior of DBSCAN was evaluated in three different configurations: using Euclidean distance, using cosine similarity, and on a representation space reduced with UMAP. In all cases, the effect of ϵ on the number of formed clusters, the proportion of points labeled as noise, and clustering quality measured by the silhouette coefficient was explored.

First, when applying DBSCAN with Euclidean distance on the VGG16 feature vectors, $\epsilon = 18$ was used, selected after exploratory analysis. This configuration generated 7 clusters and a high proportion of points marked as noise (39%). The silhouette coefficient reached a value close to 0.2

(silhouette = 0.195), suggesting a partially defined structure with some compactness within clusters but also overlaps between them. The PCA projection (Figure 3) showed coherent groupings in some regions of the space, especially for classes such as cars or airplanes. However, mixes between visually similar classes such as fruits and flowers were observed, also reflected in the confusion matrix (Figure 4). External evaluation yielded an Adjusted Rand Index (ARI) of 0.098 and a Van Dongen index of 0.492, indicating low agreement with the true labels.

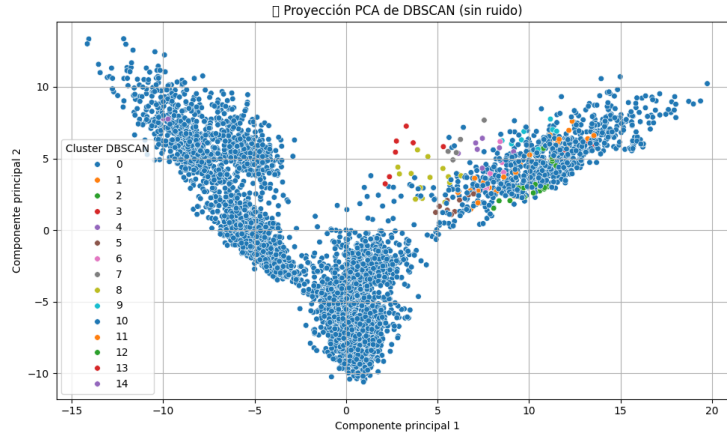


Figure 3: Two-dimensional projection using PCA of the clusters obtained with DBSCAN using Euclidean distance and $\epsilon = 18$. Each color represents a cluster assigned by the algorithm.

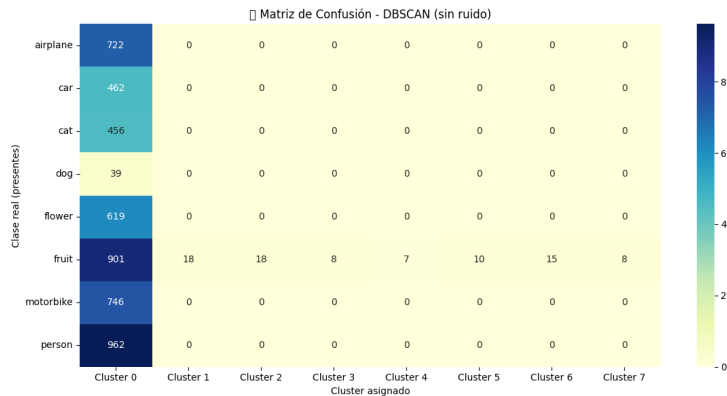


Figure 4: Confusion matrix between the true dataset labels and the clusters assigned by DBSCAN with Euclidean distance. Color intensity represents the number of elements per class-cluster combination.

The second configuration evaluated DBSCAN with cosine similarity as the distance metric. Although multiple ϵ values were explored, $\epsilon = 0,254$ was finally used, selected after exploratory analysis. With this configuration, 7 clusters were detected and 3,54% of the samples were labeled as noise (244 points). Clustering performance with respect to the true classes was evaluated using external metrics, obtaining an Adjusted Rand Index (ARI) of 0,6333 and a Van Dongen index of 0,1832, suggesting moderate correspondence with the true dataset labels.

To complement the analysis, PCA dimensionality reduction was applied (Figure 5), enabling a visual inspection of cluster distributions in the projected space. In addition, a confusion matrix (Figure 6) between true labels and DBSCAN assignments was built, identifying highly pure regions along with other clusters that grouped visually similar categories.

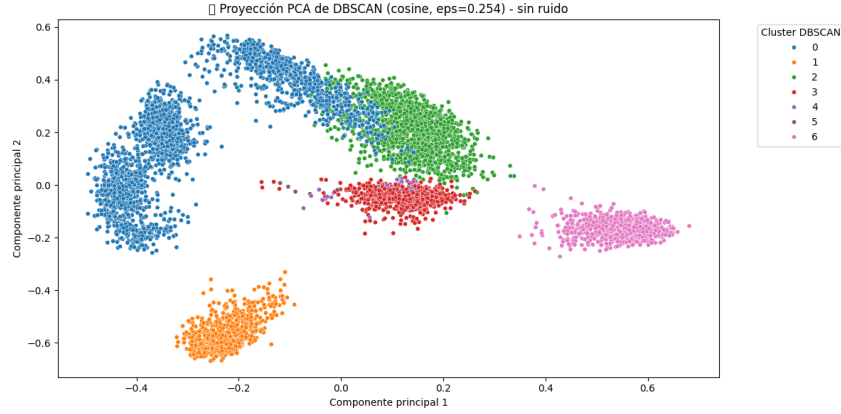


Figure 5: PCA projection of the clusters generated by DBSCAN using cosine similarity. Each color indicates a cluster assigned by the algorithm.

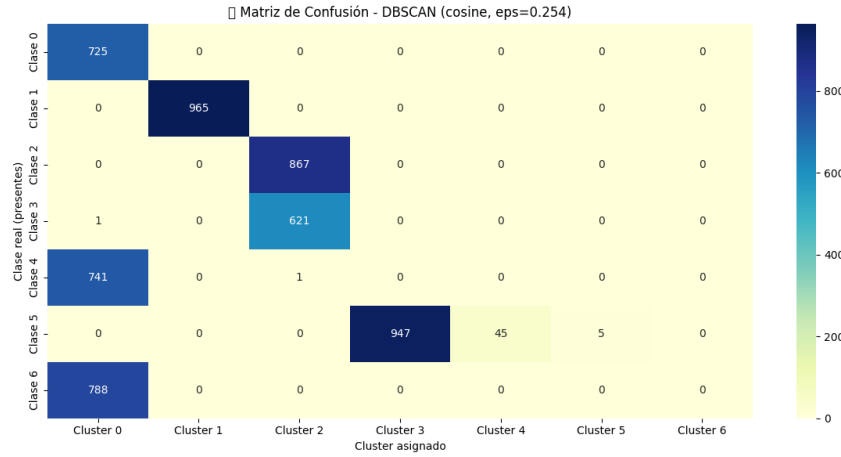


Figure 6: Confusion matrix between the true labels and the clusters obtained with DBSCAN using cosine similarity. It reflects the degree of agreement between classes and clusters.

The third configuration consisted of applying DBSCAN on variables reduced with UMAP. This technique represented the data in a two-dimensional space while preserving neighborhood relationships. Although multiple eps values were explored, eps = 1,2 and min_samples = 5 were finally used, with which 12 clusters were detected and no points were labeled as noise.

Clustering performance was evaluated using external metrics, obtaining an Adjusted Rand Index (ARI) of 0,8723 and a Van Dongen index of 0,0591, indicating high correspondence between the formed clusters and the true classes. Although the silhouette coefficient (0,81) did not improve considerably relative to previous configurations, the UMAP space facilitated the visual interpretation of the clusters.

The visual evaluation was complemented with a PCA projection (Figure 7), where a clearer segmentation was observed in several regions of the space. The confusion matrix (Figure 8) reflected this improvement, showing greater alignment between clusters and true classes. In summary, dimensionality reduction with UMAP not only facilitated visualization but, in this configuration, also produced superior results in terms of quantitative fit.

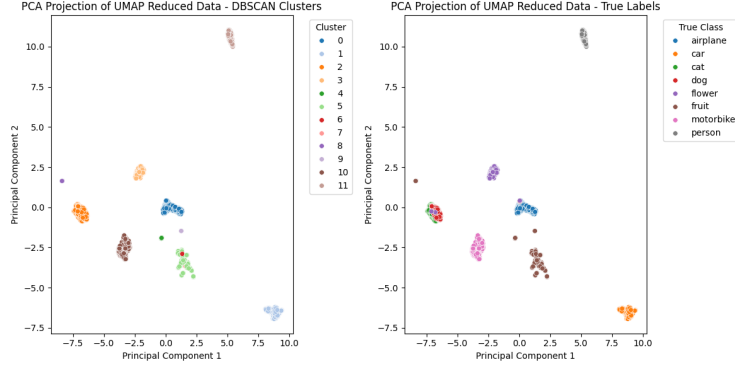


Figure 7: Visual comparison between the clusters generated by DBSCAN (left) and the true dataset labels (right), both projected in the space reduced with UMAP.

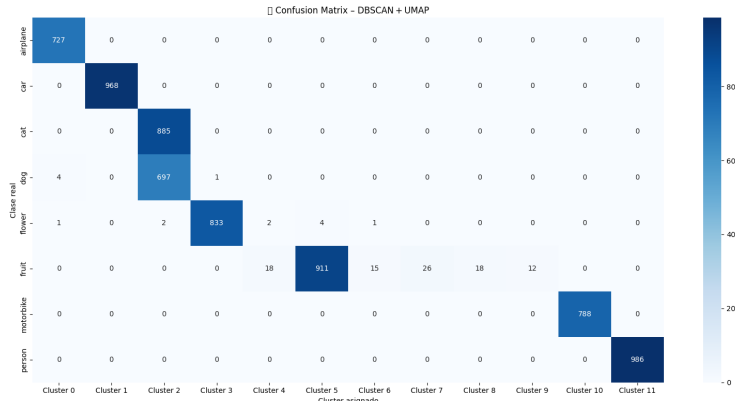


Figure 8: Confusion matrix between the true labels and the clusters obtained by DBSCAN after dimensionality reduction with UMAP. It reflects the degree of agreement between classes and clusters.

Overall, the results show that the choice of metric and representation space has a considerable impact on DBSCAN performance. The Euclidean metric enabled a reasonable segmentation with low silhouette, the cosine metric provided greater tolerance to noise but lower alignment with the classes, and the combination with UMAP facilitated visual exploration without consistently improving clustering-quality metrics.

5 Experiment 3: Hierarchical Clustering

5.1 Specific methods

For this experiment, agglomerative hierarchical clustering was applied using ward linkage and the Euclidean distance metric, directly on the feature vectors extracted with VGG16. This algorithm builds a hierarchy of successive merges of instances as a function of their proximity, without requiring prior dimensionality reduction or assumptions about cluster shape.

The optimal number of clusters was evaluated using the silhouette coefficient, testing values between 2 and 15 clusters. The maximum silhouette value was achieved with 8 clusters, matching the true number of classes in the dataset. The model was then trained with this fixed value and the results were compared with the true labels.

To evaluate performance, external validation metrics were computed, including the Adjusted Rand Index (ARI) and the Van Dongen index. A two-dimensional projection using PCA was also performed, and a confusion matrix was generated to analyze the correspondence between true classes and the resulting clusters.

5.2 Results and discussion

Hierarchical analysis was performed by applying an agglomerative clustering algorithm with ward linkage and Euclidean distance on the VGG16 representations. A dendrogram cut corresponding to 8 clusters was selected, consistent with the true number of classes in the dataset.

Regarding internal evaluation, the obtained silhouette coefficient was approximately 0,200. This value indicates intermediate separation and cohesion between clusters, suggesting that, although the groupings are reasonably compact, there is no clear geometric separation among all groups in the extracted feature space.

However, external validation results show outstanding performance. The Adjusted Rand Index (ARI) reached 0,9865, indicating an almost perfect correspondence between the clusters assigned by the algorithm and the true labels. Likewise, the Van Dongen index was 0,0065, reflecting minimal structural dissimilarity between both partitions.

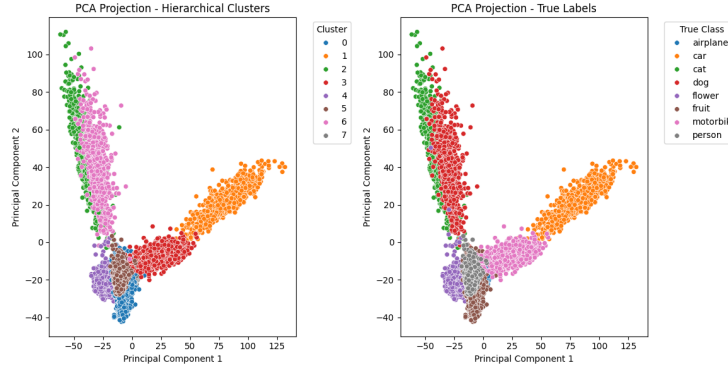


Figure 9: Two-dimensional PCA projection of the clusters generated by the agglomerative hierarchical algorithm (ward linkage). Each color represents an assigned group.

This difference between internal and external metrics suggests that, although clusters are not extremely compact in geometric terms, the algorithm achieves excellent agreement with the true classes, which indicates that VGG16 representations contain useful semantic information that is effectively captured by the hierarchical approach.

The PCA projection (Figure 9) showed a clear separation between groups, with well-defined boundaries for most classes. The confusion matrix (Figure 10) evidenced strong alignment between generated clusters and true classes, with a dominant main diagonal, indicating that most items in each class were correctly grouped.

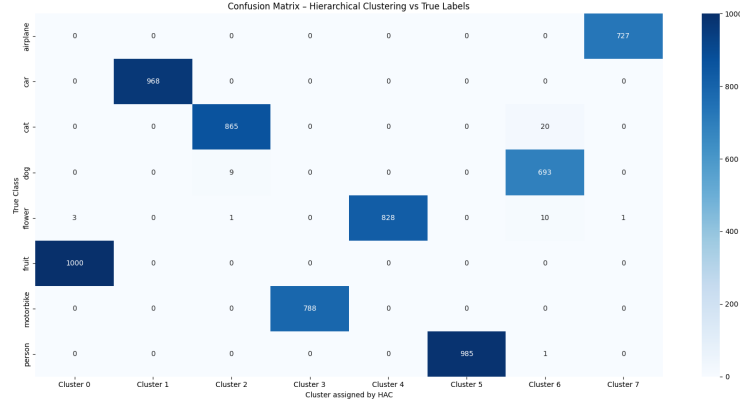


Figure 10: Confusion matrix between the true dataset labels and the clusters obtained by hierarchical clustering. The dominant diagonal indicates high agreement between classes and clusters.

Hierarchical clustering proved superior to K-Means and DBSCAN, including DBSCAN using the UMAP-based variant.

6 Automatic object detection in an image

To address the task of automatic object segmentation, an RGB image with multiple distinguishable elements was selected (Figure 11a). The image was resized to reduce its resolution and facilitate subsequent processing. It was then converted to grayscale (Figure 11b).

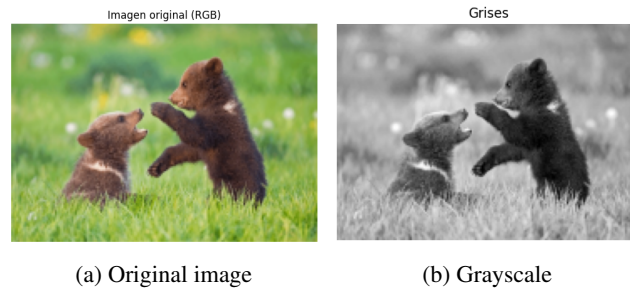


Figure 11: Original RGB image and its grayscale version

The initial segmentation was obtained using the Connected Component Labelling (CCL) algorithm, which identifies contiguous regions of connected pixels. In cases where objects appeared merged, a complementary procedure based on the Euclidean distance transform and the watershed algorithm was used to separate overlapping components.

The effect of the binarization type on the CCL result was explored. Different automatic thresholding strategies (Otsu, Yen, Li, Triangle) as well as a manually adjusted threshold (0.4) were compared (Figure 12).

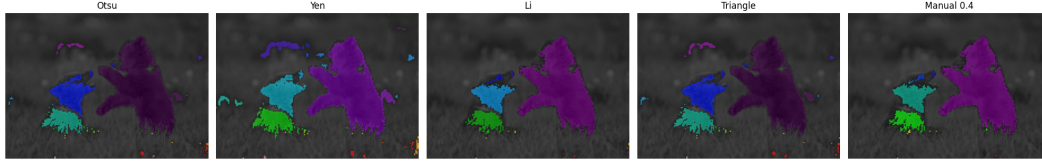


Figure 12: CCL results using different automatic binarization-threshold selection methods: Otsu, Yen, Li, Triangle, and a manual threshold (0.4). The labels are overlaid on the original grayscale image. Differences in cleanliness and number of detected regions can be observed depending on the chosen threshold.

To reduce the presence of unwanted small regions and improve the coherence of segmented areas, additional morphological operations were applied to each resulting binary mask. In particular, a morphological closing with a disk structuring element of radius 3.5 was used, followed by removal of small objects (smaller than 500 pixels) and filling of internal holes. This processing consolidated the regions corresponding to the main objects in the image, discarding irrelevant details and background noise. The final result is shown in Figure 13, where the detected objects appear clearly defined after this refinement.

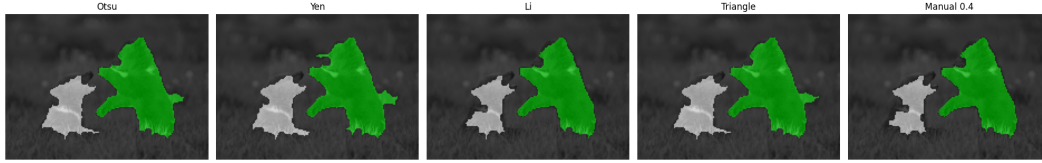


Figure 13: Segmented masks after applying morphological operations to the different binarization results. Morphological closing, small-object removal, and hole filling were used to improve the definition of the main objects and reduce background noise. The resulting regions are overlaid transparently on the original image.

After comparing different binarization methods and applying the corresponding morphological operations, Otsu's automatic threshold was selected as the best option for this case. This method achieved a good balance between complete detection of the main objects (the two bears) and a clean background segmentation, especially after morphological refinement. The combination of Otsu with morphological filters yielded connected, compact, and visually coherent regions, suitable for CCL labeling and for subsequent comparison with spectral clustering results.

In a second stage, spectral clustering was applied to the image pixels, using as input a similarity graph based on smoothed intensities. This graph models the image as a network in which nodes represent pixels and edges reflect local dissimilarity between them.

To construct the affinity matrix of the graph used in spectral clustering, a Gaussian transformation was applied to pixel-to-pixel distances. This transformation depends on the parameter β , which controls how quickly similarity decays with distance. Different β values were explored by visualizing the transformation curve for each (Figure 14). It was observed that with low β values, most edges maintain high similarities, generating a very dense graph, while high values produce a sharp drop, restricting similarity to very close neighbors.

A value of $\beta = 20$ was chosen as a compromise between local connectivity (capturing fine edges) and separation between regions. This value produced good results.

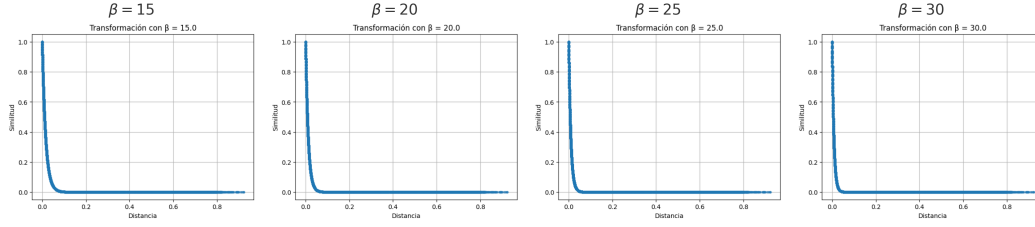


Figure 14: Distance-to-similarity transformation curves for different values of the parameter β . As β increases, similarity decays more rapidly with distance, generating a more local and selective graph. This visualization was used to analyze the model’s sensitivity to local contrast.

In addition, the two available methods to assign labels within spectral clustering were compared: *k-means* and *discretize*. Although both produced reasonable segmentations, the *discretize* method generated boundaries that better matched the visual structures of the objects, especially when regions were compact and had smooth gradients. For this reason, *discretize* was selected for the final tests.

The influence of the number of clusters k on segmentation was also evaluated. Different configurations ($k = 3$, $k = 4$, $k = 5$ and $k = 6$) were tested, visualizing the results on the original image (Figure 15). It was observed that the large bear, with darker tones, is correctly segmented starting from $k = 3$, showing good separation from the background. However, the small bear, which is lighter and shares colors with the background, is not well distinguished in any configuration: its body remains partially merged with the background or split across multiple regions. Increasing the number of clusters to $k = 5$ or $k = 6$ introduces more detail, but also more fragmentation and noise, without substantially improving segmentation of the small bear. Therefore, $k = 4$ is a reasonable compromise between separating the main object well (the large bear) and avoiding excessive over-segmentation.

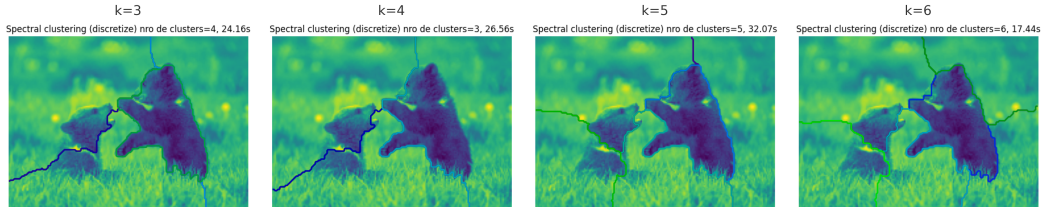


Figure 15: Spectral clustering results varying the number of clusters $k \in \{3, 4, 5, 6\}$. The large bear is correctly segmented starting from $k = 4$, while the small bear, with lighter tones, remains partially confused with the background in all cases.

The performance of spectral clustering applied only to the binary image resulting from Otsu preprocessing was also evaluated, as was done for Connected Component Labelling. In this approach, the graph was constructed considering only the pixels corresponding to the mask (foreground), using their coordinates as features. This variant reduced problem complexity and focused exclusively on the objects. The result (Figure 16) shows effective segmentation of both bears, with well-defined edges even for the small bear, which previously blended with the background. This strategy proved especially useful to complement segmentation based on the full image, reducing clustering errors in irrelevant regions.

Spectral clustering sobre imagen binaria nro de clusters=3, 66.87s

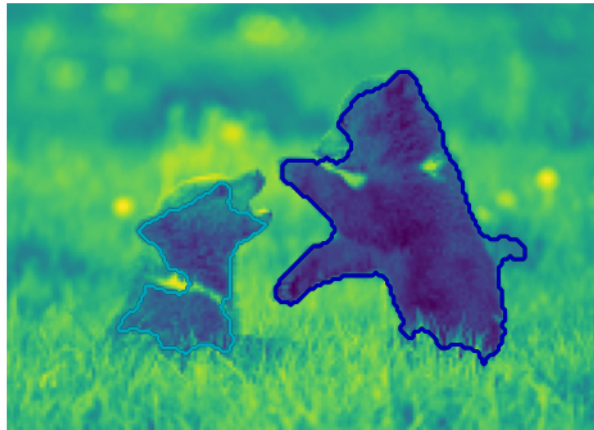


Figure 16: Result of spectral clustering applied to the binary image. Both bears were successfully separated using only the mask pixels as input.

From the obtained results, it is concluded that Connected Component Labelling (CCL) strongly depends on the quality of the initial binarization. When the binary mask correctly separates objects from the background, CCL achieves precise and efficient segmentation. However, its performance is affected if objects have grayscale levels similar to the background or if they appear partially merged. In such cases, the method may fail to properly separate regions unless complemented with additional techniques such as watershed.

On the other hand, spectral clustering allows working directly on the smoothed image without needing prior binarization. This flexibility enables it to better capture smooth transitions or complex structures. Nevertheless, in this particular case, the method had difficulty correctly segmenting objects with intensity levels similar to the background, such as the small bear, which ended up partially fused with the surroundings.

When spectral clustering was applied to the binarized image, segmentation improved significantly, with results comparable to CCL. In scenarios where objects are partially overlapping or do not separate well with a fixed threshold, the spectral approach may be more suitable, since it does not depend exclusively on binary connectivity but on more general similarity relations between pixels.

7 Conclusions

The experiments show that clustering performance strongly depends on the data representation and the chosen algorithm. K-Means produced good results for visually separable classes, while DBSCAN was sensitive to the choice of parameters and distance metrics. The combination with UMAP improved visualization and, in some cases, clustering quality. Hierarchical clustering achieved the best overall performance in terms of alignment with the true classes.

On the other hand, in the automatic segmentation task, CCL proved to be an efficient technique when objects are well separated and contrasted from the background, but it strongly depends on binarization and may fail if objects are merged or have grayscale levels similar to the environment. Spectral clustering, by contrast, enables a more flexible segmentation and can better capture complex structures, although it requires more parameter tuning. When applied to the binary image, it achieved results comparable to CCL, showing that both methods can complement each other depending on scene characteristics.

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